

Applications of Statistical Modeling:
Supplement to Module #1, Lecture #4

Software for Fitting Models to Longitudinal Data

Fall 2025

Overview

- Available software procedures in R for fitting multilevel models to normally distributed outcomes
- Software procedures in R for fitting marginal linear models to normally distributed outcomes
- Software procedures in R for fitting models to non-normal outcomes:
 - Multilevel Modeling Approaches
 - GEE Approaches
- Take-Home Points

The Autism Data Set

- An example data set created for the illustrations in presented here is available on Canvas (autism.XXX)
- This data set has socialization (i.e., VSAE) data collected over time (ages 2, 3, 5, 9, and 13) during childhood for several children diagnosed with autism
- We also have information on expressive language groups for the children at baseline
- See West et al. (2014, *Linear Mixed Models*) for additional details on this data set

Always Start with Spaghetti Plots!

- **R:** see Lecture 3 notes, use ggplot with group = ID
- For binary outcomes, it's more helpful to generate plots of proportions over time for different groups of cases using an aggregate data set of proportions for each time/group combination, and similar methods

Multilevel Models for Normal Outcomes: lme() in R nlme package

- Example syntax:

```
library(nlme)
model.fit <- lme(vsae ~ age +
  factor(sicdegp) + age:factor(sicdegp),
  random = ~ age - 1 | childid,
  method = "REML", na.action = "na.omit",
  data = autism)
summary(model.fit)
```

Multilevel Models for Normal Outcomes: lme() in R nlme package

- Notes on the syntax:
 - Note the use of (-1) in the random argument to omit the random intercepts, which would be included by default
 - An unstructured variance-covariance matrix for the random effects is also included by default
 - The lme() function is one of the few multilevel modeling functions in R that allows for correlated errors, potentially with non-constant variance across groups; see umich.edu/~bwest/chapter3_R_final.txt
 - R also uses different reference categories for categorical factors by default (see output on next slide)

Multilevel Models for Normal Outcomes: lme() in R nlme package

Random effects:

Formula: ~age - 1 | childid

age Residual

StdDev: 2.863204 9.194127

Fixed effects: vsae ~ age + factor(sicdegp) + age:factor(sicdegp)

	Value	Std.Error	DF	t-value	p-value
(Intercept)	2.481581	1.2710970	449	1.952314	0.0515
age	2.821922	0.4698435	449	6.006090	0.0000
factor(sicdegp)2	-1.293389	1.6732326	155	-0.772988	0.4407
factor(sicdegp)3	-4.230069	1.8620681	155	-2.271705	0.0245
age:factor(sicdegp)2	0.984540	0.6194261	449	1.589439	0.1127
age:factor(sicdegp)3	4.462789	0.6884915	449	6.481981	0.0000

Fitting Marginal Models to Normally Distributed Outcomes

- In most general-purpose statistical packages, marginal linear models can be fitted to normal outcomes by using (for the most part) the same multilevel modeling procedures
- We simply leave out the explicitly specified random effects, and use appropriate syntax to specify a covariance structure for the random errors
- **Remember: These are NOT mixed-effects models**
- We'll now consider the same examples with an **unstructured** covariance structure for the errors

Fitting the Marginal Model: R

- We use the `gls()` function in the `nlme` library:

```
# make ordering of observations clear
```

```
autism$index <- autism$age
```

```
autism$index[autism$age==2] <- 1
```

```
autism$index[autism$age==3] <- 2
```

```
autism$index[autism$age==5] <- 3
```

```
autism$index[autism$age==9] <- 4
```

```
autism$index[autism$age==13] <- 5
```

```
# use unstructured variance-covariance matrix for errors
```

```
marg.model.fit <- gls(vsae ~ age + factor(sicdegp) +
```

```
  age:factor(sicdegp), na.action = "na.omit",
```

```
  correlation = corSymm(form = ~index | childid),
```

```
  weights = varIdent(form = ~1 | age), autism)
```

```
summary(marg.model.fit)
```

```
getVarCov(marg.model.fit)
```

Fitting the Marginal Model: R

Coefficients:

	Value	Std.Error	t-value	p-value
(Intercept)	1.087562	1.1258657	0.965978	0.3344
age	3.015190	0.5303880	5.684876	0.0000
factor(sicdegp) 2	0.708811	1.4922022	0.475010	0.6350
factor(sicdegp) 3	-3.302054	1.6563710	-1.993547	0.0467
age:factor(sicdegp) 2	0.463266	0.7030202	0.658965	0.5102
age:factor(sicdegp) 3	4.332499	0.7781059	5.568006	0.0000

Marginal variance covariance matrix

	[,1]	[,2]	[,3]	[,4]	[,5]
[1,]	10.8080	8.8481	8.3421	24.575	43.036
[2,]	8.8481	54.5610	55.0130	94.118	187.270
[3,]	8.3421	55.0130	140.2800	193.460	368.330
[4,]	24.5750	94.1180	193.4600	811.050	1158.900
[5,]	43.0360	187.2700	368.3300	1158.900	2183.500

Standard Deviations: 3.2876 7.3865 11.844 28.479 46.728

Software for Fitting Longitudinal Models to Non-Normal Outcomes

- We'll now consider logistic models for a binary indicator of whether a child's VSAE score is above the overall median score (14)
- We'll consider multilevel (hierarchical) and GEE (marginal) approaches to fitting the logistic models
- Recall that there are many alternatives for estimation when considering the multilevel modeling approaches!

Multilevel Logistic Model: R

- **Syntax:**

```
# glmmPQL() in the nlme package uses PQL;  
# glmer() in the lme4 package uses AGQ (recommended...)  
library(lme4)  
autism$vsae_high[autism$vsae > 14] <- 1  
autism$vsae_high[autism$vsae <= 14] <- 0  
mlm <- glmer(vsae_high ~ age + factor(sicdegp) +  
  age:factor(sicdegp) + (age - 1|childid),  
  family = binomial, nAGQ = 7, data = autism)
```

Multilevel Logistic Model: R

Random effects:

Groups Name Variance Std.Dev.

childid age 0.2096 0.4578

Number of obs: 610, groups: childid, 158

Fixed effects:

	Estimate	Std. Error	z value	Pr(> z)	
(Intercept)	-4.4657	0.6936	-6.439	1.20e-10	***
age	0.5768	0.1325	4.353	1.34e-05	***
factor(sicdegp)2	1.0618	0.7197	1.475	0.140085	
factor(sicdegp)3	-0.7242	1.2420	-0.583	0.559810	
age:factor(sicdegp)2	0.1514	0.1618	0.936	0.349380	
age:factor(sicdegp)3	1.6591	0.4680	3.545	0.000393	***

Signif. codes: 0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Logistic Model using GEE: R

- Syntax:

```
# make sure data are sorted by ID and age!
library(geepack)
autism$sicdegp <- factor(autism$sicdegp)
autism$sicdegp <- relevel(autism$sicdegp, ref="3")
# for some reason, using an unstructured working
# correlation matrix causes R to crash in this case...
bingee <- geeglm(vsae_high ~ age + sicdegp + age:sicdegp,
  id=childid, waves=age, family=binomial("logit"),
  data=autism, corstr="exchangeable")
# geepack makes the anova() function available, which is
# nice for overall tests of factors / terms in the model
summary(bingee)
anova(bingee)
```

Logistic Model using GEE: R

Coefficients:

	Estimate	Std.err	Wald	Pr(> W)	
(Intercept)	-3.888	1.317	8.71	0.0032	**
age	1.663	0.510	10.62	0.0011	**
sicdegp1	1.155	1.355	0.73	0.3943	
sicdegp2	1.925	1.345	2.05	0.1522	
age:sicdegp1	-1.374	0.512	7.20	0.0073	**
age:sicdegp2	-1.327	0.513	6.70	0.0096	**

Signif. codes: 0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Estimated Correlation Parameters:

	Estimate	Std.err
alpha	0.153	0.244

Logistic Model using GEE: R

Analysis of 'Wald statistic' Table

Model: binomial, link: logit

Response: vsae_high

Terms added sequentially (first to last)

	Df	X2	P(> Chi)	
age	1	93.0	<2e-16	***
sicdegp	2	75.7	<2e-16	***
age:sicdegp	2	7.5	0.023	*

Signif. codes: 0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Take-away Points

- Remember the difference in possible inferences between multilevel and marginal models:
 - **Hierarchical / Multilevel:** include explicit random effects, and inferences are subject-specific (i.e., you are “controlling” for random subject effects, and fixed effects represent what happens when a covariate changes for a given subject)
 - **Marginal:** Population-averaged effects (what happens, on average, across all subjects in a study?)
- Also, remember that inferences are valid under a MAR assumption; include relevant covariates!

Take-away Points

- Software for fitting longitudinal models to normally distributed outcomes is fairly well-established
- Software for fitting longitudinal models to non-normal outcomes is relatively newer, and many different estimation options are possible
- Examine the sensitivity of your results to different estimation methods!
- **This supplement has provided you with “how-to” knowledge; specifying the appropriate model for a given data set is up to you!**